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Machine learning to optimize nanocomposite materials for electromagnetic interference shielding

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ABSTRACT

Carbon-based fillers/polymer nanocomposites for electromagnetic interference (EMI) shielding have attracted researchers' attention due to their excellent electrical conductivity and lightweight. The numerous material design features make it flexible to prepare required shielding composites. However, the developments of the composites usually depend on researchers' experience and repeated experiments, leading to longer development cycles and more costs. This work employs the machine learning approach to establish a shielding effectiveness rapid prediction model and analyze the critical factors and rules in material design, optimizing the new materials development and reducing experiments. Firstly, a dataset of carbon-based conductive particles/polymer nanocomposites for EMI is established, including the most accessible material and structure features. We take Weighted Average Ensemble strategy to ensemble five diverse base models on the dataset and find the final prediction model outperforms all base models. Besides, the importance of features is analyzed by variable importance rankings. The rules that critical features influence the EMI are investigated through model-agnostic techniques: partial dependence and individual conational expectation plots. The prediction model is a valuable tool to predict the shielding performance rapidly. Coupled with rules obtained, the model can guide materials development, shorten the development circus, and reduce costs.

1. Introduction

Nowadays, various electronic products make our human society convenient and efficient. However, electromagnetic radiation pollution also causes substantial economic losses every year and threatens human health [1]. Therefore, it is necessary to develop electromagnetic Interference (EMI) shielding materials to resolve the electromagnetic radiation pollution problem. Compared with traditional metal materials, conductive polymer composites have the advantages of light weight, low cost, and corrosion resistance. Especially due to the excellent mechanical and electrical properties of composites of carbon-based nanoparticles such as CNTs, graphene, and CNFs, they have been extensively studied for EMI for decades [2–14]. Many factors affect the shielding effectiveness of composites, including the selection and loading of conductive nanoparticles, processing methods, and material structure [1,15–21]. Numerous material design features make it easy to prepare suitable material for engineering or functional requirements [22–24].

However, the development of EMI composites usually relies on researchers' experience, inspiration, and tedious experiments to explore suitable design features, making development inefficient and costing a lot of time, money, and resources. Therefore, non-experimental methods to research composites for EMI have attracted researchers' attention. For example, Tunakova et al. used the coaxial transmission line method to predict the shielding effectiveness of Carbon-fiber reinforcements for epoxy composites [25]. Munalli D et al. also used the same method to predict the shielding effectiveness of laminated carbon fiber/epoxy composites [26]. In addition, similarly, Zhu et al. applied various effective medium theories to predict the effective electromagnetic properties of composites containing barium strontium titanate and/or nickel zinc ferrite inclusions [27]. However, they all belong to numerical methods, which are computationally expensive and unsuitable for rapid prediction.

With the rapid development of big data and artificial intelligence technologies, approaches of machine learning have substantially impacted the material field [28–30], including material discovery and design [31,32], properties prediction [33–43], struct testing [44–46] and so on. For example, Zheng et al. used machine learning approaches to discover the potential antifouling property of existing polymer

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brushes [47], Matsumoto et al. applied machine learning for advanced characterization and accurate performance prediction of porous material [48], Yao et al. used machine learning to classify single-layer and three-layer nanocomposite films based on SAXS features of the films [44]. The machine learning models can predict the performance of anonymous data, and also learn rules to guide the design of new materials. Zhang et al. reported an approach of machine learning to understand and optimize nanocomposite membranes for reverse osmosis [49]. Jin et al. selected carbon-based materials for tetracycline and sulfamethoxazole adsorption through machine learning [50]. Eroglu et al. assessed the critical materials and cell design factors for high-performance lithium-sulfur batteries [51].

Compared with traditional experimental methods, machine learning can significantly reduce labor and material costs and it can also speed up numerical methods by training on their results [52]. However, few reports about the machine learning approach applied to developing carbon-based particles/polymer composites for EMI. Therefore, in this work, we first establish a data set of carbon-based conductive nano-particles/polymer nanocomposites for EMI from published literature. Based on it, we train five diverse machine learning models. Then we take the strategy of Weighted Average Ensemble to ensemble them. Finally, a rapid EMI prediction model with an R^2 of 0.82 is obtained. It indicates the obtained model is a helpful tool for researchers to rapidly screen design features. We also analyze the feature importance through Drop-column feature importance and use partial dependence plots to explore how the critical features influence the EMI individually and synergistically. The rules obtained can also guide materials development.

2. Method

2.1. Data set

First, a dataset of polymer composite materials for EMI is built. The features and EMI are contained for each sample. Material features (matrix, filler, filler loading), structural features (thickness, porosity,

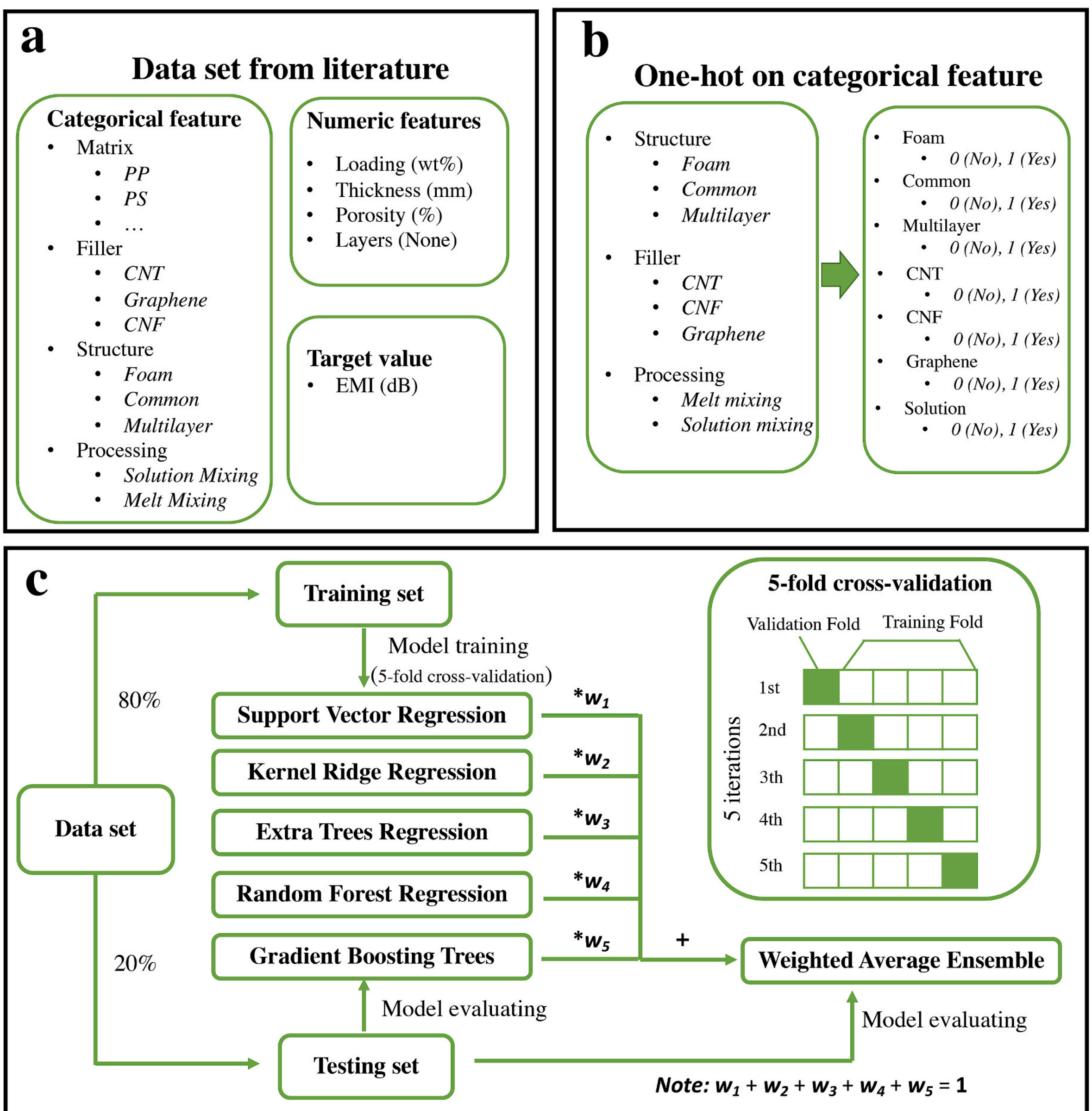

Fig. 1. The composition of data set (a); Schematic diagram of one-hot on categorical features (b); Schematic diagram of the model training process, including 5-fold cross-validation (c).

layers), and processing features (solution mixing, melt mixing, other mixing) are considered here. These features can also be categorized as categorical or numerical features, as shown in Fig. 1(a). Part of the data is directly collected from the published literature's descriptions and tables when the rest is extracted from the figures through the online tool Plot Digitizer 2.6.8 (http://plotdigitizer.sourceforge.net/). 321 samples are collected from 43 papers listed in S2 in Supporting Information.

After the data set is established, data preprocessing is performed. As shown in Fig. 1(b), in order to allow categorical features to be numerically computed during training models, the one-hot encoding method converts the categorical feature into sub-category features with "0" and "1", where "0" indicates that the sample belongs to the sub-category, and "1" indicates that the sample does not belong to the sub-category. In particular, the feature Matrix contains too many categories, and as shown in Fig. 2(d), it has no special regular to target performance, so it is dropped during model training. In addition, the feature Structure is encoded into three sub-category features Common, Foam, and Multi-layer. Because for composite materials, the feature Common is the initial default structure, and the samples will default to Common if they are not marked as a special structure. Therefore, the feature Common is a redundant feature that needs to be removed during training models. The vacancy values of one feature are filled with the means of all the data of the feature. Then the data set is randomly shuffled. According to the distribution of EMI values, the data set is divided into the training set and test set according to the ratio of 8:2. The training set is used to train the machine learning models, while the test set is as unknown samples to evaluate the models' performance. Normalization is performed on both data sets before feeding the data into the model. The tools used to process the data set are Python v3.7 and the libraries, including NumPy v1.19.2, Pandas v1.2.3, Scikit-Learn v0.24.

2.2. Model selection

In this work, we employed five diverse models to train the EMI prediction models and used them as base models to obtain an ensemble model through Weighted Average Ensemble (WAE), which outperforms all the base models. The five base models employed are Support vector machine (SVM), Kernel ridge regression (KRR), Extra Trees Regression (ETR), Random Forest Regression (RFR), and Gradient Boosting Trees (GBT). In addition, a simple multi-linear regression model, Ridge Regressor (Ridge) is also trained for comparison. The introductions of them can be found in section S1 in Supporting information.

2.2.1. Weighted average ensemble

Ensemble modeling is a method that employs multiple diverse base models to predict an output, acting and performing like a single model. It can reduce the generation error of the prediction because the various errors of the base models average out. Weighted Average Ensemble (WAE) is an ensemble modeling, expecting good models to contribute more to the prediction, and the less skillful models contribute less to the prediction. Specifically, WAE is an approach that the final prediction is calculated as the average of the weighted predictions of base models by the performance of the base models. The Expression is shown as below:

$$H(x) = \sum_{i=1}^{T} w_i \, h_i(x) \tag{6}$$

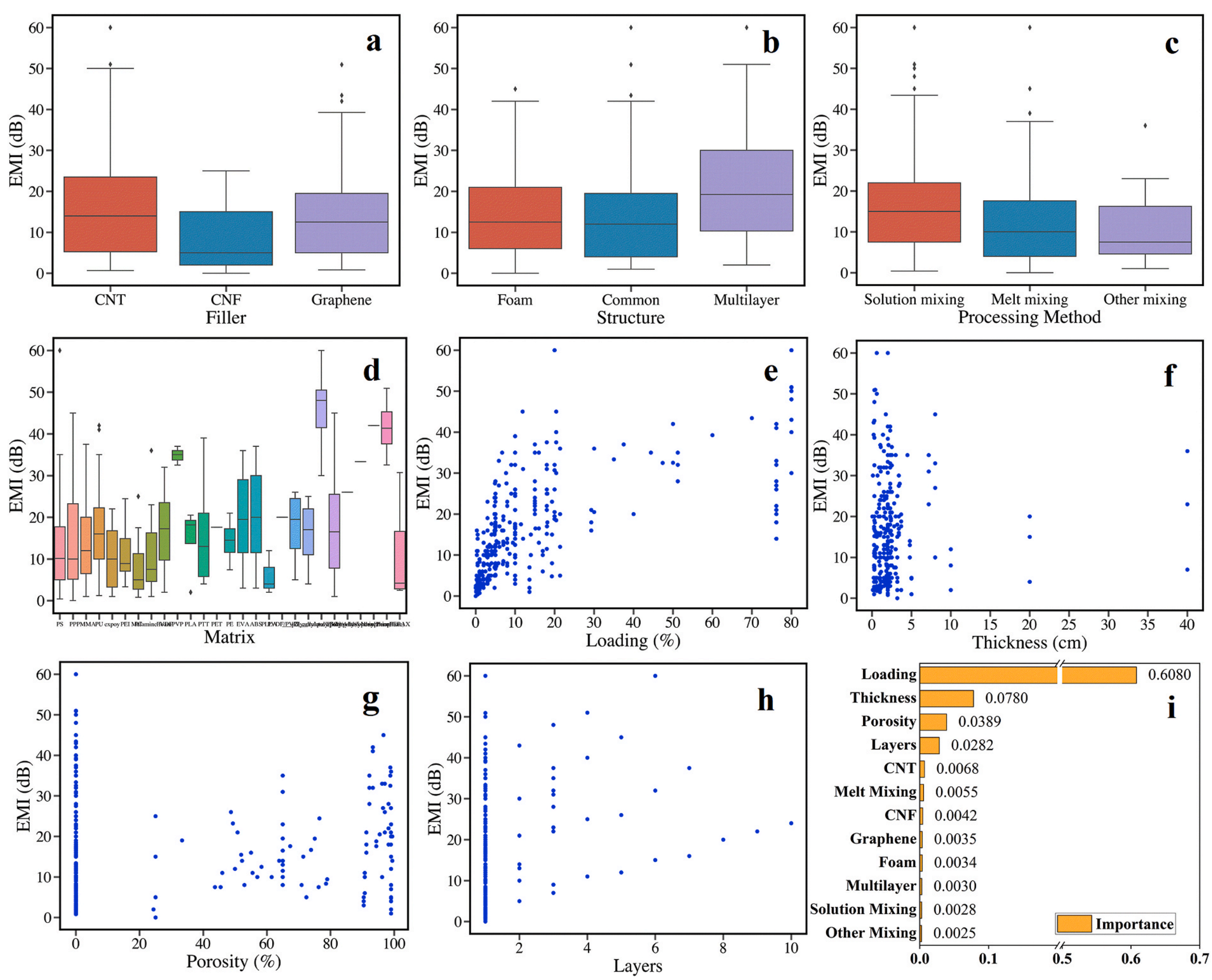

Fig. 2. Box diagrams and scatter plots of features (a–h); The feature importance ranking of the features (i).

Where H is the ensemble model; T is the number of the base models; hi is the ith base model; The model weights w_i are positive values, and the sum equals one, allowing the weights to indicate the percentage of trust or expected performance from each model h_i.

2.3. Model training

This work employs the K-fold cross-validation method to evaluate the generalization of the model more accurately during model training on the training set, as shown in Fig. 1(c). The training set is randomly divided into K groups of the same size, called folds. This work performs 5-fold with 5 iterations. For each iteration, the model is trained on 4 training folds of the training set and evaluated on the remaining validation fold. A model is obtained according to the average result of 5 evaluations on each iteration. In the selection and determination of model hyperparameters, the method of Cross grid search is also employed. The same training method was used to train all five base models. The weighted average ensemble model is obtained directly by assembling the five trained single models.

2.4. Performance evaluation metrics

After training, the model performance should be evaluated on the testing data. This work adopts three metrics to evaluate the performance of the models, including Mean Absolute Error (*MAE*), Root Mean Squared Error (*RMSE*), and R-Squared (R^2).

Mean Absolute Error (MAE):

$$MAE = \frac{1}{m}\sum_{i=1}^{m}|\widehat{y}_i - y_i| \tag{7}$$

Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{\frac{1}{m}\sum_{i=1}^{m}(\widehat{y}_i - y_i)^2} \tag{8}$$

R-Squared(R^2):

$$R^2 = 1 - \frac{\sum_{i=1}^{m}(\widehat{y}_i - y_i)^2}{\sum_{i=1}^{m}(\overline{y} - y_i)^2} \tag{9}$$

In which m indicates the sample numbers in the data set; y_i is the actual target value of the i_{th} sample, and $\widehat{y}_i$ is the i_{th} sample's prediction value, while y_{ave} is the average of all samples' actual target values.

For *MAE* and *RMSE*, the value closer to 0 indicates the numeric error between the models' prediction values and the actual values is small. The value of R^2 is between 0 and 1, and it reflects the percentage of the dependent variable variation that the model explains. Notably, in order to eliminate data distribution bias caused by dividing the data set into the training set and testing set, the data set is randomly divided 100 times. The obtained models are re-trained and re-evaluated 100 times on these data sets. The average of 100 evaluations results is used as the final evaluation result for each model.

2.5. Feature importance

Feature importance refers to techniques that assign a score to input features based on how useful they predict a target value. The score represents the importance of each feature. A higher score means that the specific feature will be more important. In this work, Drop-column feature importance is adopted. Its steps to measure feature importance is shown as follows:

1. Train WAE with all features as a baseline.
2. Measure baseline R^2 with testing set.
3. Select one feature whose importance is to be measured.
4. Retrain WAE from scratch with all features but the selected one.
5. Measure R^2 with testing set with the selected feature dropped.
6. The importance of the selected feature is the R^2 degradation from the baseline.
7. Iterate step 3. through step 6. for all features.

3. Result and discussion

3.1. Visual analysis of the data set

Before training the prediction model, we first perform a visual analysis of the data set, roughly observing the data distribution and the influence of the features on EMI. Besides, this could also help us make preprocess on the data set. The eight features are divided into two groups according to whether their data is numeric. The non-numeric features include Filler, Structure, Processing Method, and Matrix, while the numeric value features consist of Loading, Thickness, Porosity, and Layers. We make box diagrams on EMI for the non-numeric value features while scattering plots for the numeric features (Fig. 2(a–h)). As shown in Fig. 2(a), most samples with CNT and graphene share a similar EMI level, but some CNT samples tend to have higher EMI. CNF samples show a little lower EMI. We speculate it could result from the fewer amounts of samples. The box diagram of feature Structure (Fig. 2(b)) shows a similar result with the feature Filler. Multilayer structure offers a higher performance than Common and Foam, which could be due to the multilayer leading a relatively thicker sample. The feature Processing method in literature can be classified into two categories: Solution mixing and Melt mixing. As shown in Fig. 2(c), the EMI of the solution mixed samples tends to be higher. It can be explained as Solution mixing is more favorable for the dispersion of conductive filler. Fig. 2(d) shows that most of the samples with polymer matrixes maintain similar EMI. Although few matrixes offer somewhat different EMI, no conclusion can be made here. The scatter plot of the numeric features also gives some information. Fig. 2(e) indicates that the samples' EMI entirely increases while increasing the filler loading. However, as the loading exceeds about 20%, the increasing trend slows down. The scatter plot of the feature Layers (Fig. 2(h)) also offers a similar result, and the growing trend slows down as the layers arrive about 5. Fig. 2(g) indicates a roughly increasing trend when the porosity increases except for the porosity of 0.

It must be reminded here that the analysis above cannot be considered as conclusions but just an overall grasp of the data situation, which could help data preprocessing.

3.2. Evaluation of ML models

The data set is randomly divided into the training set and testing set 100 times to ensure the reliability of the models. The training and evaluation of the models with the same hyperparameters are also performed 100 times. Table 1 shows the evaluation results of the models on the training set and testing set, which are the average values of the 100 evaluations. Firstly, the simple linear model, Ridge Regressor shows very poor performance. It turns out that this work requires more advanced machine learning models. Among the five base models, the *MAE*, *RMSE*, R^2 of decision-tree-based models: RFR, ETR, and GBT are

Table 1
The evaluation results on training set and testing set.

Model	Training set			Testing set		
	MAE	RMSE	R^2	MAE	RMSE	R^2
Ridge	6.264	8.005	0.5548	6.734	8.606	0.4840
SVR	3.592	5.345	0.8024	4.376	6.111	0.7422
KRR	3.570	4.879	0.8346	4.457	6.006	0.7482
RFR	1.745	2.349	0.9617	4.302	5.816	0.7664
ETR	0.420	1.199	0.9900	4.065	5.639	0.7788
GBT	1.500	2.203	0.9662	3.898	5.319	0.8020
WAE	**1.750**	**2.460**	**0.9600**	**3.729**	**5.118**	**0.8180**

better than the nonlinear model: SVR and KRR. It shows that the decision-tree-based models can better explain the regularity of this dataset. In particular, GBT is the best single model with performances of 3.898, 5.319, and 0.8020 on *MAE*, *RMSE,* and R^2, respectively. According to the base models' performance, the weights of WAE are tuned, and WAE with wights 0.1, 0.2, 0.2, 0.2, 0.3 are found to perform best. The *MAE*, *RMSE,* and R^2 of the WAE are improved to 3.729, 5.118, and 0.8180 and better than the best base model GBT. It indicates that the weighted average ensemble strategy is successful. The results can be interpreted as follows, the decision tree-based model outperformed the nonlinear model on the whole dataset but didn't handle some data well, where the nonlinear model could. The weighted average strategy comprehensively considers the results given by the base models and gives better performance. In addition, Table 2 shows the standard deviation and variance of the 100 evaluations of the models. As shown in Table 2, except that the variance and standard deviation of *RMSE* of WAE are only slightly larger than that of GBT, and the variance and standard deviation of other performances are the best. Overall, it proves that the WAE performs more stable is on the data set.

We input the training set and testing data into the WAE to obtain the prediction values and make a scatter plot of the actual value – predicted value, as shown in Fig. 3(a and b). The diagonal dash line in the figure means the predicted values are equal to the actual values. The scatters of the data set close to the diagonal dash line also indicate that model performs a good prediction. The linear fitting line with a slight angle with the diagonal line also supports the model's good performance. These results indicate that WAE is a useful tool. The researchers can use WAE to predict the shielding effectiveness rapidly and screen potential good design features. It will significantly reduce the cost of experimental the cost of experiments.

3.3. Feature importance analysis

The ranking of feature importance is shown in Fig. 2(i). From the ranking, we can find that the importance of the feature Loading is overwhelming among the features, indicating that the loading of fillers mostly affects the shielding effectiveness. The other numeric features of Thickness, Porosity, and Layers are the following three most important features. In addition, the importance of the four numeric features above is much greater than those of the categorical features. This result can allow researchers to understand which features may more affect the shielding effectiveness and which features should be paid more attention to when designing materials.

3.4. Features dependence analysis

Based on the feature importance ranking, the most important numeric features: Loading, Thickness, Porosity, and Layers, are be analyzed how they affect the shielding effectiveness. The model-agnostic techniques: Partial dependence and individual conational expectation plots are used here.

The partial dependence plot shows the marginal effect features have on the prediction values of the machine learning model [53]. The partial dependence function for regression is as below:

$$\widehat{f}_{x_S}(x_S) = E_{x_C}\left[\widehat{f}(x_S,\ x_C)\right] = \int \widehat{f}(x_S,\ x_C)d\mathbb{P}(x_C) \tag{10}$$

In which, the x_S are the features that the partial dependence function analyzes, and the xc are the other features of the machine learning model *f*. The feature vectors x_S and x_C combined make up the total feature space *x*.

Partial dependence works by marginalizing the machine learning model output over the distribution of the features in set C, so that the function shows the relationship between the features in set S investigated and the prediction values. In general, there are only one or two features in the set S for the readability of the plots, respectively called univariate PDP and bivariate PDP.

In the actual calculation, the partial function $\widehat{f}_{x_S}(x_S)$is estimated by calculating averages in the training data, also known as the Monte Carlo method:

$$\widehat{f}_{x_S}(x_S) = \frac{1}{n}\sum\nolimits_{i=1}^{n}\widehat{f}\left(x_S,\ x_C^{(i)}\right) \tag{11}$$

In which, $x_C^{(i)}$indicates the actual feature values from the dataset for the features.

The partial function tells us for given values of features S what the average marginal effect on the prediction is. In this formula, $x_C^{(i)}$ are actual feature values from the dataset for the features not analyzed, and n is the number of samples in the dataset.

The partial dependence plot for the average effect of a feature is a global method because it does not focus on specific instances but an overall average. The equivalent to a PDP for individual data instances is called an individual conditional expectation plot [54].

The partial dependence plot for the average effect of features is a global method because it focuses on an overall average. On the contrary, the individual conditional expectation (ICE) plots to display one line per sample, showing how the prediction value of the sample changes when a feature changes. The ICE can be the plot in the same chart with univariate PDP for features investigated.

3.4.1. Univariate PDP and ICE analysis

In Fig. 3(c–f), the bold dash curves are the PDP curves on the features, and solid curves are the ICE curves. First, the graphs of the four variables are compared. The ICE curves of the feature Loading (Fig. 3(c)) are relatively clustered. It can be considered that because the impact of Loading on EMI is far more significant than others, the curves are less affected by other features. For the same reason, the other features' ICE curves are widely distributed on the figures due to different filler loadings.

Then the figure of each feature is analyzed separately. The PDP curve of the feature Loading shows an upward trend with increasing loading, but the movement slows down as the loading is around 20%. It is consistent with the data visualization analysis. Once the loading reaches a critical point, conductive channel forms inside the materials, and the EMI increases rapidly, but the trend slows down with further increasing loading. It can inspire researchers to consider other methods to improve the shielding effectiveness after the filler loading reaches a certain level. Besides, although the ICE curves of the feature Loading are not the same in value, they maintain a similar increasing trend, resulting from the influence of other features. The PDP cure of the feature Thickness (Fig. 3(d)) shows an upward trend with the increasing thickness. However, the ICE curves some different information. The materials with low loading should be thicker to increase the EMI obviously, while the materials with high loading begin to maintain good EMI from a smaller thickness. It can inspire researchers that with higher filler loading, increasing thickness is more efficient to improve the shielding effectiveness.

The PDP curve of the feature Porosity (Fig. 3(e)) shows that the EMI

Table 2
The standard deviation (std) and variance (var) of the 100 evaluations on testing set.

Model	*MAE*		*RMSE*		R^2	
	std	var	std	var	std	var
Ridge	0.5241	0.2746	0.9401	0.8837	0.1043	0.0109
SVR	0.4548	0.2069	0.6946	0.4825	0.0477	0.0023
KRR	0.4781	0.2286	0.9318	0.8683	0.0765	0.0058
RFR	0.4282	0.1833	0.7518	0.5651	0.0410	0.0017
ETR	0.4193	0.1758	0.7150	0.5112	0.0467	0.0022
GBT	0.4313	0.1860	0.6786	0.4605	0.0471	0.0022
WAE	**0.4064**	**0.1652**	**0.6865**	**0.4713**	**0.0378**	**0.0014**

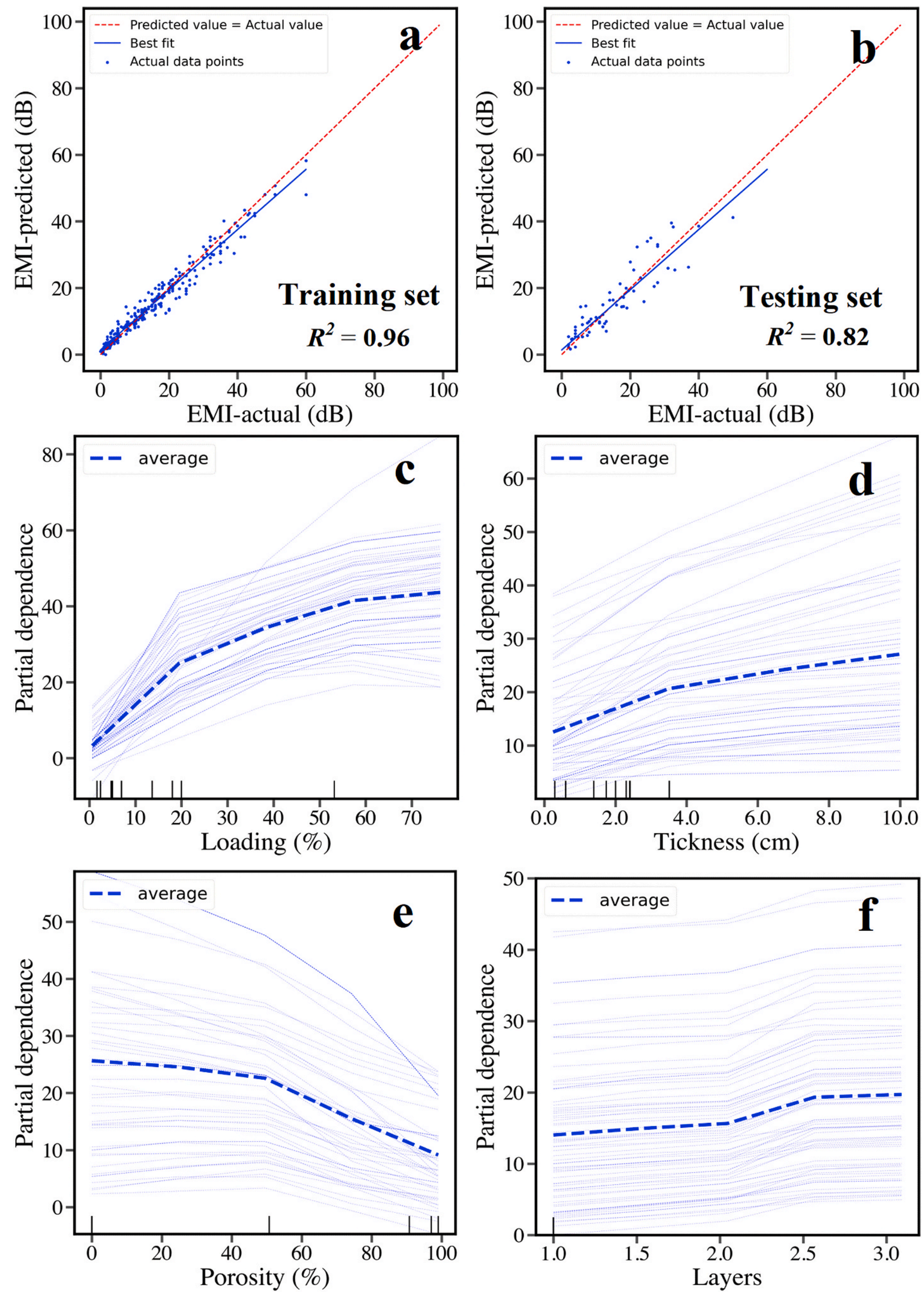

Fig. 3. Comparisons between EMI-actual and EMI-predicted on Training set (a) and Testing set (b); Univariate partial dependence plots of the features (c–f).

increases slowly until the porosity reaches about 50%, then decreases gradually with the porosity further going up. However, the ICE shows different results and can be divided into three groups. The trends of curves with moderate loadings are like the PDP curve, but the curves with low loadings increase before loading reaches 50% and then decreases. Moreover, the curves with high loadings start to go down with the porosity just presented. Combining the above results, we considered that the porosity hurts the EMI while just considering this single feature. However, for low filler loadings, the increasing porosity could contribute to the formation of the conductive channel, leading to EMI increasing. But once the porosity reaches a certain level, the conductivity increases hardly. The negative effect of the porosity works. Moreover, the porosity will reduce the surface conductivity of the entire material, decreasing the EMI. For moderate filler loadings, it can be considered that the negative and positive effects of the porosity reach a balance when the porosity is relatively low. Still, the increasing porosity will eventually cause the loss of EMI. For high filler loadings, only the negative effects work due to the already formed conductive channels. The above analysis about porosity can inspire researchers to pay attention to the balance between porosity and the shielding effectiveness while designing lightweight porous structures. The PDP curves and ICE curves of the feature Layers (Fig. 3(f)) show a stepwise improvement in changing from one layer to three layers, indicating that the multilayer structure is an effective strategy to improve EMI.

3.4.2. *Bivariate partial dependence plots analysis*

The Bivariate partial dependence plots analyze the marginal effect two features have on the EMI. As shown in Fig. 4, The plots of Loading-Thickness (Fig. 4(a)), Loading-Porosity (Fig. 4(b)), and Loading-Layers (Fig. 4(c)) all show that the gradient of the partial dependence on the feature Loading is much bigger, indicating that the feature Loading dominates the effect on the EMI. Moreover, the plot of Loading-

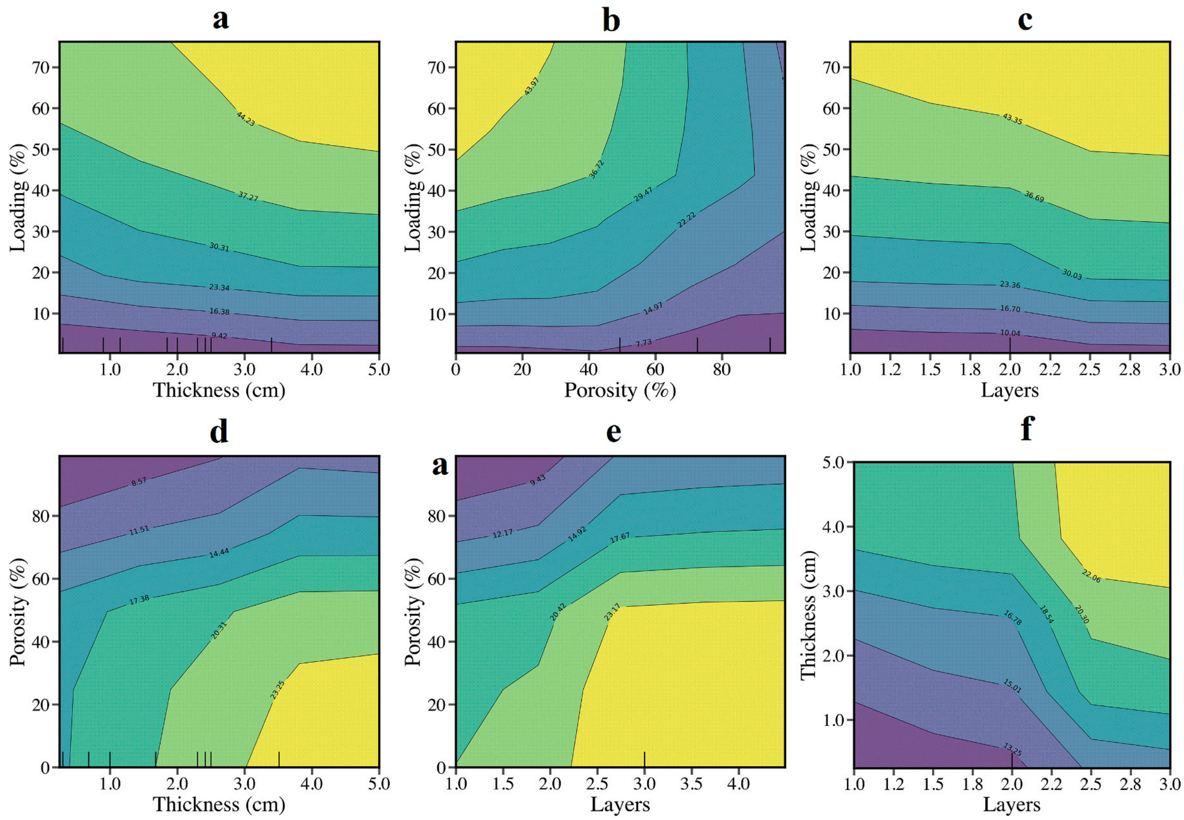

Fig. 4. Bivariate partial dependence plots of the features.

Thickness (Fig. 4(a)) shows that the EMI increases faster with the increasing thickness at higher loading. The plot of Loading-Porosity (Fig. 4(b)) shows the two opposite effects of the feature Porosity. The plot of Loading-Layers (Fig. 4(c)) shows a stepwise improvement in changing from one layer to three layers. These are all consistent with the univariate PDP and ICE analysis. In addition, the plot of Porosity-Thickness (Fig. 4(d)) and Porosity-Layers (Fig. 4(e)) shows that after the porosity exceeds a certain level, the shielding effectiveness drops quickly and can be hardly improved through increasing thickness and layers. It also indicates the importance of balancing porosity and the shielding effectiveness. And according to the plot of Thickness-Layers (Fig. 4(f)), the increase of both features leads to better EMI.

4. Conclusion

We successfully built a rapid EMI prediction machine learning model for carbon-based nanoparticles composites, with an R^2 of 0.82. It indicates the obtained model is a helpful tool for researchers to rapidly screen design features. Besides, the feature importance ranking gives the most critical features: filler loading, thickness, porosity, and the number of layers, to which researchers should pay more attention. Moreover, several guidelines for EMI nanocomposites design can be derived from the PDPs and ICPs analysis results. 1. After the filler loading reaches a certain level, researchers should consider other methods to improve the shielding effectiveness; 2. With higher Loading, increasing thickness is more efficient to improve the shielding effectiveness; 3. High porosity can seriously impair the shielding effectiveness, and researchers need to balance its relationship with the shielding effectiveness as researchers design porous EMI nanocomposites; 4. Multilayer is an effective strategy to improve the shielding effectiveness.

In addition, this work also provides a relatively complete process for building composite machining models and how to analyze the model to obtain guidance for composites design. This approach can also be applied to composite materials in other fields.

Author statement

This work has not been submitted elsewhere for publication, in whole or in part, and all the authors listed have approved the manuscript that is enclosed. And we declare there is no conflict of interest.

Author contributions

Meng Shi: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing - Original Draft, Investigation, Visualization; Chang-Ping Feng: Investigation, Writing – Review & Editing. Jiang Li: Supervision, Project administration. Shaoyun Guo: Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compscitech.2022.109414.

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